
Practice Paper 1

Fundamentals of ML, Data Preparation & Feature Engineering,
Supervised Learning

50 questions. Five options each (A-E). One correct answer per question. Answer key at the end.

Name: _____

Student No.: _____

Date: _____

Score: _____

1 A digital lender defines: Task T = flag a transaction as fraudulent at initiation; Experience E = eighteen months of transaction logs with labels arriving weeks late from chargebacks; Performance P = cost-weighted recall at a fixed daily alert budget. Under Mitchell's operational definition of learning, which statement about this instantiation is correct?

- A. The definition is invalid because E must consist of fully labelled data available at training time, not delayed labels
- B. P is mis-specified; only accuracy satisfies Mitchell's definition of a performance measure
- C. The definition is valid, but E and P already encode significant engineering reality: delayed, noisy labels and an operations-constrained metric
- D. T is invalid because a task must be defined independently of the deployment moment, not "at the moment it is initiated"
- E. Mitchell's definition requires E , T and P to be drawn from the same distribution, which is violated here

2 The expected risk $R(f) = \mathbb{E}_{(x,y) \sim D}[L(y, f(x))]$ cannot be computed directly in practice. Why, and what do we compute instead?

- A. Because L is non-convex; we instead compute a convexified surrogate risk over the same distribution D
- B. Because D is unknown; we instead compute the empirical risk over a finite training sample drawn i.i.d. from D
- C. Because x and y are not independent; we instead compute a conditional risk given x
- D. Because f is not differentiable; we instead compute a subgradient approximation of $R(f)$
- E. Because the loss function L is unbounded; we instead compute a clipped empirical loss

3 A lender's default model was trained pre-pandemic. After a macroeconomic shock, applicants with an identical transaction profile now default at a substantially higher rate, while the demographic mix of applicants is unchanged. This is best classified as:

- A. Label-prior shift only, since $P(y)$ has changed while $P(x)$ and $P(y|x)$ are both fixed
- B. Data drift (covariate shift), since the shock altered the input distribution $P(x)$
- C. Concept drift, since $P(y|x)$ has changed while the applicant profile distribution $P(x)$ is essentially unchanged
- D. Sampling bias, since the training sample no longer represents the population of interest
- E. Training/serving skew, since the feature computation pipeline itself has changed

- 4** According to the course's "build-or-not" framework, which scenario most strongly argues for a hand-written rule rather than an ML model?
- A. Fraud patterns that shift on a weekly basis, defeating any static rulebook
 - B. A high-dimensional, strongly non-linear relationship between hundreds of interacting behavioural features
 - C. A short, auditable rule already meets the required standard and errors must be fully explainable
 - D. The organisation wants to extract latent structure such as customer segments from unlabelled data
 - E. The environment is expected to drift continuously as adversaries adapt
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- 5** For a continuous target, the constant prediction that minimises expected squared error and the constant prediction that minimises expected absolute error are, respectively:
- A. The conditional mean and the conditional mean
 - B. The conditional mode and the conditional median
 - C. The conditional mean and the conditional median
 - D. The conditional median and the conditional mean
 - E. The conditional variance and the conditional mean
-
- 6** A portfolio has a 3% default rate. A model that predicts "repays" for every applicant achieves roughly 97% accuracy while catching zero defaulters. This scenario is used in the course notes to illustrate that:
- A. Accuracy is undefined under class imbalance and cannot legally be reported
 - B. A model must exceed 97% accuracy before accuracy even begins to carry decision-relevant information, since it weighs all errors equally under asymmetric costs
 - C. The correct fix is always to oversample the minority class before any model is trained
 - D. Precision and recall are mathematically undefined when a class has zero true positives
 - E. ROC-AUC and accuracy are always equal under class imbalance, so either may be reported
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- 7** In the end-to-end ML lifecycle presented as a loop rather than a waterfall, which claim about where effort is spent is asserted by the course notes?
- A. Roughly 80% of real project effort lives in framing and data work (steps 1-3), not in model selection
 - B. Model selection and hyperparameter tuning dominate the effort budget of real-world projects, mirroring the emphasis of most textbooks
 - C. Deployment and monitoring together consume the majority of project time in a typical first iteration
 - D. Framing the problem is a one-time, ten-minute exercise that should not be revisited once modelling begins
 - E. The loop terminates once a model beats its baseline on the validation set; monitoring is optional afterwards
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8 Data leakage is formally defined in terms of a mismatch between two feature maps. Which pair, and what does the mismatch mean?

- A. ϕ_{train} and ϕ_{serve} ; leakage exists when the model is evaluated on $\phi_{\text{train}}(x)$ but will be applied to a $\phi_{\text{serve}}(x)$ that differs in distribution
- B. ϕ_{public} and ϕ_{private} ; leakage exists whenever a private feature is exposed in a public API response
- C. ϕ_{raw} and ϕ_{scaled} ; leakage exists whenever any scaling is applied before model training
- D. ϕ_{train} and ϕ_{test} ; leakage exists only when the same row literally appears in both splits
- E. ϕ_{linear} and $\phi_{\text{nonlinear}}$; leakage exists whenever a nonlinear transform is fitted on the full dataset

9 A lender considers simply removing gender and region from the feature set to “achieve fairness.” Which principle from the course’s fairness discussion is most directly violated by assuming this is sufficient?

- A. Group-fairness criteria are provably always mutually satisfiable once protected attributes are removed
- B. Removing the protected attribute does not, by itself, produce fairness, because other features can act as proxies for it
- C. Explainability requirements are automatically satisfied once protected attributes are dropped
- D. Fairness auditing is unnecessary once the model has been dropped from production monitoring
- E. Dropping a feature always increases model accuracy on the protected subgroup

10 A lender sets cost of a missed default at KES 10,000 and cost of a wrongly declined good borrower at KES 3,000, with correct decisions costing nothing. Using the cost-optimal threshold rule, at what calibrated probability t^* should the lender begin flagging an applicant as high risk?

- A. $t^* \approx 0.500$
- B. $t^* \approx 0.769$
- C. $t^* \approx 0.231$
- D. $t^* \approx 0.300$
- E. $t^* \approx 0.100$

11 Under Rubin’s taxonomy, a lender notices GPS coordinates are missing exclusively for feature-phone users, and this is fully explained by the observed device-type column. This missingness pattern is:

- A. MNAR, because the missingness depends on the unobserved GPS value itself
 - B. MCAR, because the missing rows form a uniform random sample
 - C. MAR, because conditional on an observed column (device type), missingness carries no further information about the missing GPS values
 - D. Ignorable only if the sample size exceeds 10,000 rows
 - E. Structurally undefined, since Rubin’s framework only applies to numeric, not geospatial, features
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12 An analyst finds that among applicants with GPS present, the 90-day default rate is 4.1%, while among those with GPS missing, it is 9.7%. Per the course’s diagnostic protocol, the correct response is to:

- A. Discard the GPS column entirely, since missing data can never be predictive
 - B. Impute the missing GPS values using the global mean latitude and longitude
 - C. Conclude the missingness is MCAR and proceed with simple mean imputation
 - D. Retain a binary missingness indicator as a feature in its own right, since the gap suggests MNAR-like structure where missingness itself is signal
 - E. Apply z-score outlier removal to the GPS column before imputing
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13 Why does the course recommend the median, not the mean, for simple imputation of transaction-amount columns?

- A. The median is always unbiased under MNAR, while the mean is never unbiased under any missingness regime
 - B. Transaction amounts are heavy-tailed, so a single very large value can drag the mean far from a typical value, while the median is robust to that tail
 - C. scikit-learn’s SimpleImputer only supports the median strategy for numeric columns
 - D. The mean requires the data to be normally distributed, which is a strict prerequisite for any imputation
 - E. The median is computed on the full dataset (train and test together), avoiding leakage, whereas the mean cannot be
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14 For a high-cardinality categorical column, why does the course recommend an explicit “UNKNOWN” sentinel over most-frequent-category imputation for missing categorical values?

- A. Most-frequent imputation is computationally infeasible for more than 50 categories
 - B. A sentinel preserves the information that the value was absent rather than silently inflating the modal category’s apparent frequency
 - C. scikit-learn’s OneHotEncoder rejects most-frequent imputation for categorical columns
 - D. A sentinel guarantees the encoder will never encounter an unseen category at serving time
 - E. Most-frequent imputation always introduces target leakage, while a sentinel never does
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15 Scikit-learn's `IterativeImputer`, used to approximate MICE, models each incomplete column as a function of the other columns and cycles until the fills stabilise. Which statement about it is accurate?

- A. It uses a random forest regressor by default and cannot express fill uncertainty
- B. It uses a Bayesian ridge regressor by default; with `sample_posterior=True`, fills become draws rather than point estimates, propagating imputation uncertainty
- C. It is guaranteed to converge to the unique MCAR-optimal solution within a fixed number of steps set by `max_iter`
- D. It requires no `random_state`, since the underlying regressor is deterministic
- E. It should always be preferred over simple imputation because it never degrades serving-time latency

16 A feature has quartiles $Q1 = 20$ and $Q3 = 80$ ($IQR = 60$). Under the standard IQR outlier rule used in the notes, a point with value 185 is:

- A. Flagged as an outlier, since the upper fence is $Q3 + 1.5 \cdot IQR = 80 + 90 = 170$, and 185 exceeds it
- B. Not an outlier, since the upper fence is 260
- C. An outlier only if a z-score confirms $|z| > 3$ as well
- D. Not an outlier, since the IQR rule only applies to the lower tail for right-skewed monetary data
- E. Undeterminable without knowing the mean and standard deviation

17 The Isolation Forest anomaly score is defined as $s(x) = 2^{(-E[h(x)]/c(n))}$, where $h(x)$ is the average isolation depth over trees and $c(n)$ normalises for the expected depth of a binary search tree on n points. A score close to 1 indicates:

- A. A typical, well-connected inlier that requires many splits to isolate
- B. An anomaly, since it was isolated in unusually few splits relative to the expected depth
- C. A tie between the Gini and entropy impurity criteria at that point
- D. That the point lies exactly on the K-means cluster boundary
- E. Numerical instability in the isolation forest's random seed

18 A cleaning script winsorizes or deletes a small cluster of applicants whose accounts show three enormous round-figure deposits shortly before applying, a known balance-inflation gaming pattern. Per the course's "deleting the signal" pitfall, this is:

- A. Correct practice, since any value beyond three IQRs should always be removed for model stability
- B. A mistake if applied mechanically, because these are exactly the behavioural extremes the model is meant to detect, not data-quality errors
- C. Irrelevant, since gaming patterns cannot be captured by any tabular feature set
- D. Correct only when the deposits exceed the 99.9th percentile of the full unsplit dataset
- E. Acceptable, provided the deletion happens after model training rather than before

19 Why must entity resolution (merging records for the same real-world customer across multiple SIM cards) happen before the train/test split, and splitting must then use the resolved customer_id?

- A. Because scikit-learn's train_test_split function raises an error on duplicated names
- B. Because failing to do so scatters copies of the same customer across train and test, letting the model "recognise" the test customer from training rather than generalise
- C. Because entity resolution is only mathematically valid on data structured by SIM identifier
- D. Because fuzzy name matching requires a fixed train/test split to compute similarity scores
- E. Because regulatory rules in Kenya require deduplication only for the test set

20 Which scaler is most fragile to a single extreme outlier, because that one point redefines the scaling bounds and crushes the rest of the distribution toward zero?

- A. The robust scaler, since it uses the interquartile range
- B. Standardization (z-score scaling), since it uses the sample mean
- C. Min-max scaling, since a single outlier redefines x_max or x_min
- D. Quantile transformation, since it maps values through the empirical CDF
- E. Box-Cox transformation, since λ is estimated by maximum likelihood

21 Net cash-flow features in a mobile-money dataset can be negative. Which transform is used specifically because it extends the Box-Cox idea to handle the full real line, including negative values?

- A. The $\log(1+x)$ transform
- B. The Yeo-Johnson transform
- C. Min-max scaling
- D. The hashing trick
- E. Standardization

22 With an intercept term, one-hot encoding all K levels of a nominal feature (rather than K-1) creates a problem for an unregularised linear model. This problem is:

- A. The "dummy trap": the K indicator columns sum to one, producing perfect collinearity with the intercept
- B. Catastrophic forgetting, since the model overwrites earlier learned weights
- C. The curse of dimensionality, which only affects distance-based learners, not linear ones
- D. Vanishing gradients in the coefficient estimation
- E. A violation of Rubin's MAR assumption

23 The hashing trick maps a categorical level c to a bucket via $\phi(c) = h(c) \bmod d$. Which property makes it attractive for an agent-identifier column in a system where new agents are onboarded continuously?

- A. It guarantees zero collisions regardless of d , unlike frequency encoding
- B. It requires no fitted vocabulary at all, so levels unseen at training time are still mapped to a valid bucket at serving time
- C. It always outperforms target encoding in predictive lift
- D. It automatically satisfies the MAR assumption for the encoded column
- E. It eliminates the need for any train/test split discipline

24 Target encoding is defined as $TE(c) = (n_c \cdot \bar{y}_c + m \cdot \bar{y}) / (n_c + m)$. For a category with $n_c = 4$, $\bar{y}_c = 0.75$, global mean $\bar{y} = 0.05$, and smoothing constant $m = 20$, the encoded value is closest to:

- A. 0.750
- B. 0.050
- C. 0.400
- D. 0.167
- E. 0.283

25 Why does naive (non-out-of-fold) target encoding constitute self-leakage, and what is the standard fix?

- A. Because each row's encoded value is computed partly from its own label; the fix is to encode each fold using statistics fitted only on the other folds
- B. Because target encoding always violates GDPR; the fix is to use one-hot encoding instead
- C. Because target encoding cannot represent unseen categories; the fix is to use the hashing trick instead
- D. Because target encoding requires a neural network to compute; the fix is gradient clipping
- E. Because target encoding is only valid for continuous targets; the fix is bucketing the target first

26 For a periodic feature such as hour-of-day, why must both $\sin(2\pi t/T)$ and $\cos(2\pi t/T)$ be included rather than just one?

- A. Because scikit-learn's ColumnTransformer requires an even number of derived features
- B. Because either coordinate alone maps two genuinely different times of day to the same encoded value, destroying information
- C. Because $\sin$ and $\cos$ together are required to satisfy the MCAR assumption
- D. Because a single trigonometric feature always produces negative variance
- E. Because tree-based models cannot split on real-valued features at all

27 In scorecard practice, the weight of evidence for a bin is $\text{WoE}(\text{bin}) = \ln[P(\text{bin} \mid y=0) / P(\text{bin} \mid y=1)]$. A logistic regression fitted on WoE-coded bins is described in the notes as being equivalent to:

- A. A Random Forest with monotonic constraints
 - B. The traditional credit scorecard format that many East African lenders' risk committees expect
 - C. A kernel SVM with an RBF kernel
 - D. An unregularised neural network with a single hidden layer
 - E. A Gaussian mixture model with two components
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28 Huyen's field guide enumerates six common causes of leakage. Which of the following is explicitly listed as one of the six?

- A. Using more than fifteen engineered features in a single model
 - B. Choosing a tree-based model over a linear model for tabular data
 - C. Scaling or imputing using statistics computed on the full dataset before splitting
 - D. Reporting PR-AUC instead of ROC-AUC on an imbalanced dataset
 - E. Training a model for more than 100 boosting rounds
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29 In the temporal evaluation design for the credit scorer, every observation carries two clocks: the feature window ends at scoring time t , and the label matures over $[t, t+90 \text{ days}]$. A loan disbursed 40 days before a split boundary should be:

- A. Included as "non-default", since 40 days is close enough to the 90-day window to be treated as resolved
 - B. Excluded from the split, since its label window has not closed and including it as non-default injects optimism
 - C. Included as "default", to bias the model conservatively toward caution
 - D. Duplicated across both the training and validation sets to increase statistical power
 - E. Used only for scaling and imputation statistics, never for the model target
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30 Logistic regression models the log-odds as a linear function: $\log[\hat{p}/(1-\hat{p})] = \theta^T x + b$. If the fitted, standardised coefficient on `tenure_months` is $\theta_j = -0.35$, one standard deviation of additional tenure multiplies the odds of default by approximately:

- A. 1.42
 - B. 0.35
 - C. 0.70
 - D. -0.35
 - E. 0.50
-

31 In scikit-learn's LogisticRegression, the regularisation strength parameter C is related to the penalty weight λ by:

- A. $C = \lambda$, so a larger C means stronger regularisation
 - B. $C = 1/\lambda$, so a larger C means weaker regularisation
 - C. $C = \lambda^2$, so C and regularisation strength move in the same direction
 - D. C has no relationship to λ ; it only controls the solver's convergence tolerance
 - E. $C = -\lambda$, so negative C values indicate strong regularisation
-

32 The soft-margin SVM objective is $\min (1/2)\|w\|^2 + C \cdot \sum_i \max(0, 1 - y_i(w^T x_i + b))$. Which statement about this objective is correct?

- A. Every training point contributes equally to the final decision boundary, regardless of its margin
 - B. Only points on or inside the margin (the support vectors) carry non-zero weight in the solution; the rest could be deleted without moving the boundary
 - C. The hinge loss term is zero everywhere except at points exactly on the boundary
 - D. Increasing C always widens the margin regardless of the data
 - E. The objective is non-convex, so multiple restarts are required to find a global optimum
-

33 Why are k-nearest-neighbours classifiers described as having a latency profile that is "the exact opposite of what production wants"?

- A. Because kNN requires GPU acceleration unavailable in most serving environments
 - B. Because there is no training phase, but each prediction costs $O(m)$ distance computations against the training set, unlike models with cheap prediction and expensive training
 - C. Because kNN can only be implemented in a non-differentiable programming language
 - D. Because kNN requires the entire training set to be re-shuffled before every prediction
 - E. Because kNN's memory footprint grows quadratically with the number of features regardless of sample size
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34 For a positive class with true recall $r = 0.7$ estimated from a validation set containing $n_+ = 150$ positives, the standard error of the recall estimate, $se(\text{recall}) = \sqrt{r(1-r)/n_+}$, is closest to:

- A. 0.0037
 - B. 0.700
 - C. 0.037
 - D. 0.21
 - E. 0.0007
-

- 35** Stratified k-fold cross-validation, rather than plain k-fold, is described as mandatory under class imbalance because:
- A. It guarantees the model achieves higher accuracy on the minority class
 - B. An unstratified fold may contain almost no positives at low prevalence, which makes the standard-error of any recall-type estimate explode
 - C. It removes the need for a held-out test set entirely
 - D. It automatically corrects for data drift between folds
 - E. It converts the classification problem into a regression problem
-
- 36** Nested cross-validation separates two roles that a single cross-validation loop conflates. What is the key distinction?
- A. An inner loop tunes hyperparameters within each outer training set; the outer loop evaluates the whole selection procedure on data it never touched, avoiding the optimism of tuning and reporting on the same folds
 - B. An inner loop trains on the test set; the outer loop trains on the training set
 - C. Nested CV eliminates the need for a validation set by using bootstrap resampling exclusively
 - D. The inner loop always uses stratified folds, while the outer loop always uses random folds
 - E. Nested CV multiplies compute by a factor of two, regardless of the number of inner folds
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- 37** Under Proposition 3.2, AUC has a probabilistic interpretation. If s_+ is the score of a random positive and s_- the score of an independent random negative, AUC equals:
- A. $P(s_+ < s_-)$
 - B. $P(s_+ > s_-) + (1/2) \cdot P(s_+ = s_-)$
 - C. $P(s_+ = s_-)$ only
 - D. The correlation coefficient between s_+ and s_-
 - E. $1 - P(s_+ > s_-)$
-
- 38** Under heavy class imbalance (e.g. 3% prevalence), which statement about ROC-AUC versus PR-AUC is correct, per the course's rule of thumb?
- A. ROC-AUC and PR-AUC are always numerically identical under any prevalence
 - B. ROC-AUC can look deceptively good because its false-positive-rate denominator is dominated by the huge negative class, while PR-AUC (whose chance baseline equals the prevalence) exposes model differences more clearly among rare positives
 - C. PR-AUC should never be reported when positives are rare, since its chance baseline is undefined
 - D. ROC-AUC's chance baseline shifts with prevalence, while PR-AUC's chance baseline is fixed at 0.5
 - E. Under imbalance, accuracy becomes a better summary statistic than either curve
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- 39** A score function $\hat{p}$ is "calibrated" if, per Definition 3.6:
- A. It ranks positives above negatives with probability 1
 - B. For every q in $[0,1]$, among all cases scored q , about a fraction q are actually positive
 - C. Its Brier score is exactly zero
 - D. It is monotonically increasing in every input feature
 - E. Its ROC-AUC exceeds 0.9
-
- 40** Per the rule of thumb from Niculescu-Mizil and Caruana cited in the notes, which pairing is most accurate regarding out-of-the-box calibration?
- A. Logistic regression is usually well calibrated out of the box; SVMs and boosted trees typically are not and benefit from post-hoc calibration
 - B. Boosted trees are always perfectly calibrated because their leaves output empirical frequencies
 - C. SVMs are inherently probabilistic and need no calibration
 - D. Logistic regression requires isotonic regression before its scores can be trusted at all
 - E. Calibration quality is unrelated to model family and depends only on dataset size
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- 41** SMOTE creates a synthetic minority point as $x_{\text{new}} = x_i + \lambda(x_{z_i} - x_i)$ with $\lambda \sim U(0,1)$. The course identifies a "cardinal sin" regarding when this must run. What is it?
- A. SMOTE must run after fitting the final model, never before
 - B. SMOTE must run inside each cross-validation fold, on the training portion only; running it before the split lets synthetic points derived from validation-fold originals leak into training
 - C. SMOTE must always use $k=1$ nearest neighbour to avoid interpolation artefacts
 - D. SMOTE must never be combined with class weighting under any circumstances
 - E. SMOTE must be applied to the test set to match the training class balance
-
- 42** McNemar's test for comparing two classifiers on one test set uses the discordant counts n_{01} and n_{10} in the statistic $\chi^2 = (|n_{01} - n_{10}| - 1)^2 / (n_{01} + n_{10})$. This test is preferred in the notes because:
- A. It requires no assumption about the distribution of scores, unlike the paired-fold approach, and was found among the best-behaved classical comparison tests
 - B. It is the only test that can be applied when sample sizes exceed 10,000
 - C. It replaces the need for a held-out test set entirely
 - D. It directly estimates the cost-optimal decision threshold
 - E. It is equivalent to bootstrapping the ROC-AUC statistic
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43 In a decision tree split search, a node holds 1,000 loans with 40 defaults ($p = 0.04$, $\text{Gini} = 2p(1-p) = 0.0768$). Splitting on "days_past_due > 7" sends 900 rows left (18 defaults, $\text{Gini} = 0.0392$) and 100 rows right (22 defaults, $\text{Gini} = 0.3432$). The improvement in impurity ΔI from this split is closest to:

- A. 0.3432
 - B. 0.0072
 - C. 0.0768
 - D. 0.0696
 - E. 0.4200
-

44 A single decision tree is described as having low bias but high variance, because greedy hierarchical growth means a small perturbation to the training data can rewrite every decision beneath a flipped root split. Which pair of remedies does the chapter present as the two structural responses to this instability?

- A. Regularisation and cross-validation
 - B. Bagging/Random Forests (parallel averaging over perturbed data) and boosting (sequential fitting of residual errors)
 - C. Standardisation and one-hot encoding
 - D. Increasing the learning rate and reducing tree depth simultaneously
 - E. Kernel methods and dimensionality reduction
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45 Under the Condorcet jury theorem (ML form), $M = 1,000$ independent classifiers, each correct with probability $p = 0.51$, vote by majority. The probability the majority vote is correct is approximately:

- A. 0.51
 - B. 0.99
 - C. 0.73
 - D. 0.50
 - E. 1.00
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46 Per Proposition 4.2, the variance of an average of M estimators with common variance σ^2 and common pairwise correlation ρ is $\text{Var}[\bar{f}(x)] = \rho\sigma^2 + (1-\rho)\sigma^2/M$. As $M \rightarrow \infty$ while ρ stays fixed and positive, this variance:

- A. Converges to zero regardless of ρ
 - B. Converges to $\rho\sigma^2$, a floor that no amount of averaging can remove
 - C. Diverges to infinity
 - D. Converges to σ^2/M , independent of ρ
 - E. Becomes negative, which is why ρ must be re-estimated
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47 Bagging trains M base learners on bootstrap samples drawn with replacement, each of size n . As $n \rightarrow \infty$, the probability that a given training row is left "out-of-bag" for a given tree converges to:

- A. $1/2$
 - B. $1/n$
 - C. e^{-1} , approximately 36.8%
 - D. $1 - e^{-1}$, approximately 63.2%
 - E. 0, since bootstrap sampling always includes every row at least once
-

48 A Random Forest restricts the split search at every node to a fresh random subset of k out of d features, in contrast to bagging alone. What problem does this specifically address?

- A. It reduces the training set size to speed up fitting
 - B. It decorrelates the trees: without it, a dominant predictor is chosen near the root of nearly every tree, keeping the pairwise correlation ρ (and hence the variance floor) high
 - C. It converts the ensemble from a variance-reduction method into a bias-reduction method
 - D. It guarantees every tree in the forest is identical, improving reproducibility
 - E. It eliminates the need for out-of-bag evaluation
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49 In AdaBoost, a weak learner achieves weighted error ε_m on its stage. Under the coefficient formula $\alpha_m = (1/2)\ln[(1-\varepsilon_m)/\varepsilon_m]$, a learner with $\varepsilon_m = 0.5$ (a coin flip) receives:

- A. $\alpha_m = 1$, the maximum possible weight
 - B. $\alpha_m = 0$, so it is effectively silenced in the final vote
 - C. A negative α_m , so the committee listens to the opposite of what it says
 - D. $\alpha_m = \infty$, since the formula is undefined at $\varepsilon_m = 0.5$
 - E. α_m equal to the average of all prior learners' coefficients
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50 LightGBM grows trees leaf-wise (best-first), splitting the leaf with the largest gain anywhere in the tree, while XGBoost's default grows level-wise. What is the direct consequence of LightGBM's strategy noted in the text?

- A. Leaf-wise growth is always slower than level-wise growth on wide tables
 - B. For a fixed leaf budget, leaf-wise growth reaches lower training loss but correspondingly overfits faster, which is why `num_leaves` is capped directly rather than left uncontrolled
 - C. Leaf-wise growth cannot handle categorical features natively, unlike level-wise growth
 - D. Leaf-wise growth guarantees a shallower tree than level-wise growth for the same number of leaves
 - E. Leaf-wise growth eliminates the need for any learning-rate hyperparameter
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Answer Key

Correct answers and brief rationale for every question.

1. **C** E and P already encode delayed labels and a cost-weighted, operations-bound metric; that is the point of the example.
2. **B** D is unknown, so we substitute the empirical risk over a finite i.i.d. sample.
3. **C** same input profile, different outcome, is concept drift ($P(y|x)$ changed).
4. **C** a short auditable rule meeting the bar is the textbook "prefer rules" case.
5. **C** squared error is minimised by the mean; absolute error by the median.
6. **B** accuracy weighs all errors equally and is misleading under 12x asymmetric costs.
7. **A** the course states 80% of effort lives in framing and data (steps 1-3).
8. **A** leakage is exactly a ϕ_{train} vs ϕ_{serve} distributional mismatch.
9. **B** dropping the attribute does not remove proxy signal correlated with it.
10. **C** $t^* = cFP/(cFP+cFN) = 3000/13000 \approx 0.231$.
11. **C** missingness fully explained by an observed column is the definition of MAR.
12. **D** the gap (4.1% vs 9.7%) is the MNAR screen; keep the indicator.
13. **B** median is robust to the heavy right tail; a single whale distorts the mean.
14. **B** a sentinel keeps the "absence" information distinct from the modal class.
15. **B** Bayesian ridge is the default regressor; `sample_posterior=True` draws fills.
16. **A** upper fence = $80 + 1.5(60) = 170$; 185 exceeds it, so it is flagged.
17. **B** scores near 1 mean short average isolation depth, i.e. an anomaly.
18. **B** these are gaming behaviour, the signal the model is meant to catch, not noise.
19. **B** unresolved entities scattered across splits let the model "recognise" test rows.
20. **C** min-max scaling is redefined entirely by a single new max or min.
21. **B** Yeo-Johnson extends Box-Cox to the full real line, including negatives.
22. **A** K one-hot columns sum to 1, producing collinearity with an intercept.
23. **B** hashing needs no fitted vocabulary, so unseen levels still map somewhere.
24. **D** $TE = (4 \times 0.75 + 20 \times 0.05)/(4+20) = (3+1)/24 \approx 0.167$.
25. **A** the row's own label contaminates its own encoded value unless folds are held out.
26. **B** a single trig coordinate maps two different times onto the same value.
27. **B** WoE-coded logistic regression is exactly the traditional scorecard.
28. **C** full-dataset scaling/imputation before splitting is cause #2/#3 on Huyen's list.
29. **B** an immature label must be excluded, not assumed non-default.
30. **C** $e^{(-0.35)} \approx 0.70$, a 30% odds reduction.
31. **B** sklearn parameterises $C = 1/\lambda$; larger C means weaker regularisation.

32. **B** only support vectors (zero or positive hinge loss) determine the boundary.
33. **B** kNN has no training cost but $O(m)$ cost per prediction, the wrong profile for low-latency serving.
34. **C** $\sqrt{0.7 \times 0.3 / 150} \approx \sqrt{0.0014} \approx 0.037$.
35. **B** unstratified folds can starve of positives at low prevalence, inflating variance.
36. **A** inner loop tunes, outer loop gives an honest, untouched-fold estimate.
37. **B** this is the exact statement of Proposition 3.2.
38. **B** ROC's FPR denominator is swamped by negatives; PR-AUC is more informative under rarity.
39. **B** this is the literal definition of calibration.
40. **A** logistic regression is well calibrated by construction; SVMs/boosted trees are not.
41. **B** SMOTE run before splitting leaks synthetic points derived from validation originals.
42. **A** McNemar's test is distribution-light on discordant pairs and well-behaved per Dietterich.
43. **B** $0.9(0.0392) + 0.1(0.3432) = 0.0696$; $\Delta I = 0.0768 - 0.0696 = 0.0072$.
44. **B** bagging attacks variance in parallel; boosting attacks bias sequentially.
45. **C** with mean 510, $sd \approx 15.8$, $P(\text{majority correct}) \approx 0.73$ as stated in the worked example.
46. **B** the $\rho\sigma^2$ term is a floor independent of M ; only the second term shrinks with M .
47. **C** $(1 - 1/n)^n \rightarrow e^{-1} \approx 0.368$ as $n \rightarrow \infty$.
48. **B** random feature subsets at each split decorrelate trees that would otherwise share a dominant root split.
49. **B** at $\varepsilon_m = 0.5$, $\ln(1)=0$, so $\alpha_m = 0$ and the learner is silenced.
50. **B** leaf-wise growth reaches lower training loss per leaf budget but overfits faster, hence capping `num_leaves`.